Bert Collins

Dr. Wang

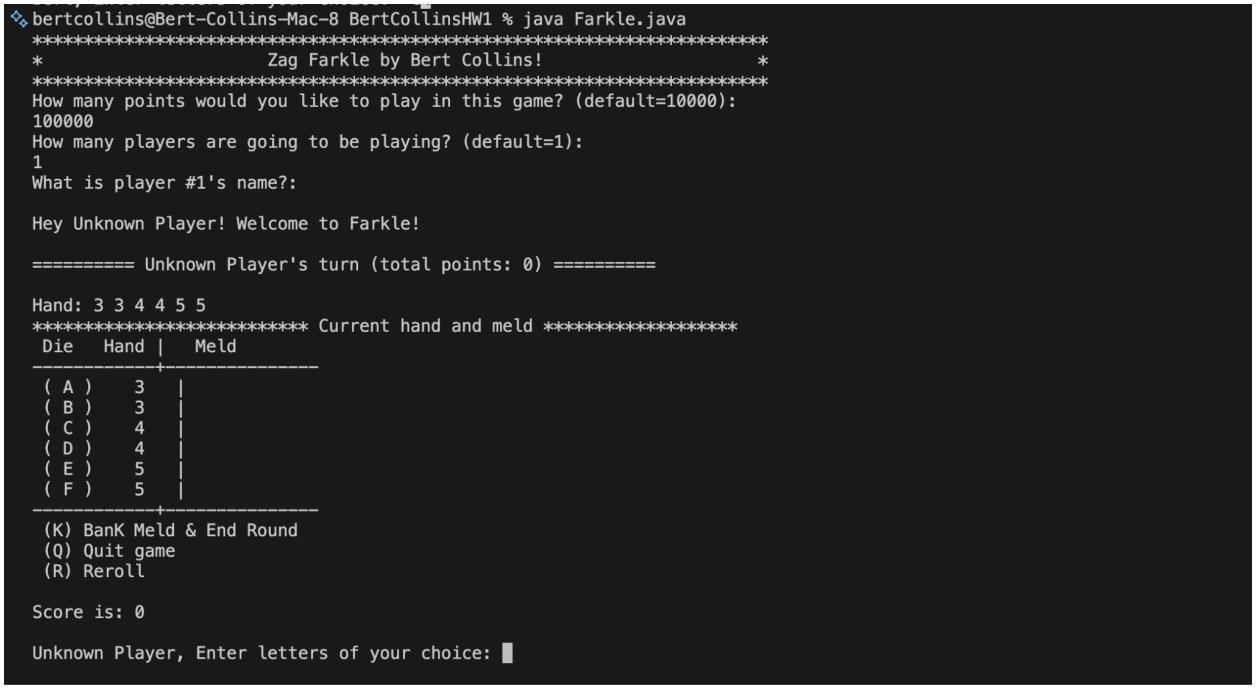
CPSC-224-01

8 November 2026

For this assignment I continued with my Farkle implementation. Previously, the game ended as soon as the player banked, but now the game is able to have more than one player play, and have each of them take turns playing until a final score is reached. A new Full House combo was also added, where it is worth 1500 points, and proper endgame logic where everyone else gets one more turn after someone reaches the target. The program still starts in Farkle.java, where an instance of the StartGame class is called, which starts the game. StartGame now asks for the target points, the number of players, and their names, then has each player take a turn using playTurn(), which prints all the players scores after every turn with a winner banner at the end. I also ended up deciding to add a Player class, which holds each player's name and individual score as private variables that can only be accessed through the following implemented public functions. DiceWork still handles shuffling the dice, changing the meld, printing the table, rerolling, detecting a farkle or hot hand, and computing the score, and also now detects a full house using the new helper function checkFullHouse() (as a mentioned before). The tests in TestCases can still be enabled by toggling the test variable in Farkle.java.

Design issues I came across were fixing a farkle bug from HW-2 and figuring out how to solve and implement multiplayer in my game. My old isFarkle function didn't work properly for specific cases, so I decided to rework it to how it originally was, with it being just 1 line of code, which in my surprise, actually worked without issues. The multiplayer side also stumped me for

a while. Even though I understood the concept that I needed to use and implement a player class, I had trouble figuring out how to create all the different turns with the logic I had implemented already, it felt like I needed to rewrite my entire code just to factor in a multiplayer function. Luckily I realized I could use my turn loop logic as its own individual function instead, where I could keep calling in another loop to be able to illustrate each term, which is what I ended up doing to solve this problem.



```
Bert Collins, Enter letters of your choice: ABCDE

Hand: 0 0 0 0 0 3
*************************** Current hand and meld ********************
 Die   Hand |   Meld
------------+---------------
 ( A )      |     2
 ( B )      |     2
 ( C )      |     2
 ( D )      |     6
 ( E )      |     6
 ( F )    3 |
------------+---------------
 (K) BanK Meld & End Round
 (Q) Quit game
 (R) Reroll

Score is: 1500

Bert Collins, Enter letters of your choice:
```

```
Hand: 0 0 0 0 0 0
************************** Current hand and meld ******************
 Die   Hand |   Meld
------------+--------------
 ( A )      |     1
 ( B )      |     2
 ( C )      |     3
 ( D )      |     4
 ( E )      |     5
 ( F )      |     6
------------+--------------
 (K) BanK Meld & End Round
 (Q) Quit game
 (R) Reroll

Score is: 1000

Bert, Enter letters of your choice: K
Turn over. Bert banked 1000 points

Current total points:
    Bert       1000

***** ENDGAME! Bert reached 100 points! *****
Everyone else gets one more turn to beat them.
*************************************************************************
*  GAME OVER! Bert wins with 1000 points!
*************************************************************************

Current total points:
    Bert       1000
```

```
bertcollins@Bert-Collins-Mac-8 BertCollinsHW1 % java Farkle.java
********************************************************************************
How many points would you like to play in this game? (default=10000):
500
How many players are going to be playing? (default=1):
3
What is player #1's name?:
Bert
What is player #2's name?:
Yin
What is player #3's name?:
Logan
Hey Bert, Yin, Logan! Welcome to Farkle!

========== Bert's turn (total points: 0) ==========

Hand: 5 3 3 1 5 6
**************************** Current hand and meld ******************
 Die   Hand |   Meld
------------+---------------
 ( A )    5   |
 ( B )    3   |
 ( C )    3   |
 ( D )    1   |
 ( E )    5   |
 ( F )    6   |
------------+---------------
 (K) BanK Meld & End Round
 (Q) Quit game
 (R) Reroll

Score is: 0

Bert, Enter letters of your choice:
```

```
 Current total points:
     Bert       750
     Yin        150
     Logan      0


 ========== Logan's turn (total points: 0) ==========

 Hand: 5 1 4 4 1 2
 **************************** Current hand and meld ******************
  Die   Hand |   Meld
 ------------+---------------
  ( A )    5   |
  ( B )    1   |
  ( C )    4   |
  ( D )    4   |
  ( E )    1   |
  ( F )    2   |
 ------------+---------------
  (K) BanK Meld & End Round
  (Q) Quit game
  (R) Reroll

 Score is: 0

 Logan, Enter letters of your choice:
```

```
  Current total points:
      Bert       750
      Yin        150
      Logan      0

  ********************************************************************************
  *  GAME OVER! Bert wins with 750 points!
  ********************************************************************************

  Current total points:
      Bert       750
      Yin        150
      Logan      0

bertcollins@Bert-Collins-Mac-8 BertCollinsHW1 %
```

